

The overall goal in the whole mini-project is to create and work with a medical database in several incremental steps. The first step is to identify the information that is going to be represented as well as special requirements. Then, the information is modeled in an ER diagram and mapped to a relational schema. The database schema is then refined and normalized. Later on, database dumps will be provided and queried.

Preliminary Database Modeling

The database should store information about drugs, their categories, the diseases they can treat, their side effects, their commercial products, their interactions with other drugs, and possibly information about related clinical trials. In general, you may use the WebMD website (<https://www.webmd.com/>) for inspiration. But be aware that WebMD stores a large variety of information and that some information there is quite verbose. You should not try to incorporate all the information available in WebMD into your database design! For most aspects, it will be sufficient to have 5 attributes instead of 20. Please choose carefully.

Sample Information

To provide you with an idea of how the information to be loaded in the database might look like, consider the sample information available at **the given excel file**. Please notice that the sample information does not correspond to what would be a database table (or relation).

Sample Queries

Your database model should allow for answering queries such as:

- a) Find the number of drugs that have nausea as a side effect
- b) Find the drugs that interact with butabarbital
- c) Find the drugs with side effects cough and headache
- d) Find the drugs that can be used to treat endocrine diseases
- e) Find the most common treatment for immunological diseases that have not been used for hematological diseases
- f) Find the diseases that can be treated with hydrocortisone but not with etanercept

Report

Please work in groups and solve the following tasks based on your current knowledge. You might want to make some assumptions about the mini-world, please clearly state them as part of your report.

1. Document and describe your database in the way you would do it (or have done it) in a project report.
2. Provide a list of tables with attributes that you would use to store your data.
3. In case you already have a good command of SQL, add the SQL commands that create your database.

The results of these tasks will serve as a basis for reflections in upcoming self studies, i.e., some upcoming tasks will ask you to compare your solution to the preliminary model that you have developed in this self study.

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